

Sisteme de ec. dif. lin. de ord. I cu coef. const.

1. $\begin{cases} x' = -5x + 2y \\ y' = x - 6y \end{cases}$, $x(0) = 3$, $y(0) = 0$

$A = \begin{pmatrix} -5 & 2 \\ 1 & -6 \end{pmatrix}$ $\quad P_A(\lambda) = \begin{vmatrix} -5-\lambda & 2 \\ 1 & -6-\lambda \end{vmatrix} = (5+\lambda)(6+\lambda) - 2 =$

$= 30 + 5\lambda + 6\lambda + \lambda^2 - 2 = \lambda^2 + 11\lambda + 28 = (\lambda+7)(\lambda+4)$

Pentru $\lambda_1 = -7 \Rightarrow \begin{cases} -5x + 2y = -7x \\ x - 6y = -7y \end{cases} \Rightarrow x = -y \Rightarrow V_{\lambda_1} = Sp\{\begin{pmatrix} 1 \\ -1 \end{pmatrix}\}$

$v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

Pentru $\lambda_2 = -4 \Rightarrow \begin{cases} -5x + 2y = -4x \\ x - 6y = -4y \end{cases} \Rightarrow x = 2y \Rightarrow V_{\lambda_2} = Sp\{\begin{pmatrix} 2 \\ 1 \end{pmatrix}\}$

$v_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$\Rightarrow$ Soluția generală a sistemului este $X = C_1 e^{\lambda_1 t} v_1 + C_2 e^{\lambda_2 t} v_2$

$= C_1 \cdot e^{-7t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + C_2 e^{-4t} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} C_1 e^{-7t} + 2C_2 e^{-4t} \\ -C_1 e^{-7t} + C_2 e^{-4t} \end{pmatrix} \Rightarrow$

$\begin{cases} x(0) = 3 \\ y(0) = 0 \end{cases} \Rightarrow \begin{cases} C_1 + 2C_2 = 3 \\ -C_1 + C_2 = 0 \end{cases} \Rightarrow 3C_2 = 3 \Rightarrow C_2 = 1 \Rightarrow C_1 = 1$

$\Rightarrow$ Soluția sistemului este: $\begin{cases} x(t) = e^{-7t} + 2e^{-4t} \\ y(t) = -e^{-7t} + e^{-4t} \end{cases}$

Deci soluţia generală a sistemului de ecuaţii diferenţiale $\mathbf{X}' = A\mathbf{X}$ va fi

$$\mathbf{X}(t) = P \begin{pmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \underbrace{C_1 e^{\lambda_1 t} \mathbf{v}_1}_{\mathbf{X}_1(t)} + \underbrace{C_2 e^{\lambda_2 t} \mathbf{v}_2}_{\mathbf{X}_2(t)}$$

↳ caz omogen, A diagonalizabilă

(b) $\begin{cases} x' = 5x - y \\ y' = x + 3y \end{cases}$, $x(0) = 0$, $y(0) = 1$

$A = \begin{pmatrix} 5 & -1 \\ 1 & 3 \end{pmatrix}$ $\quad P_A(\lambda) = \begin{vmatrix} 5-\lambda & -1 \\ 1 & 3-\lambda \end{vmatrix} = (5-\lambda)(3-\lambda) + 1 = 15 - 5\lambda -$

$3\lambda + \lambda^2 + 1 = \lambda^2 - 8\lambda + 16 = (\lambda-4)^2 \Rightarrow$

$\lambda = 4$ este valoare proprie dublă

Pentru $\lambda=4 \Rightarrow \begin{cases} 5x-y=4x \\ x+3y=4y \end{cases} \Rightarrow x=y \Rightarrow V_\lambda = Sp\{\begin{pmatrix}1\\1\end{pmatrix}\} \Rightarrow$

A nu este diagonalizabilă. $\Rightarrow$

Considerăm $v_1=\begin{pmatrix}1\\1\end{pmatrix} \in V_\lambda$ și $v_2=\begin{pmatrix}1\\0\end{pmatrix} \Rightarrow$ Soluția generală a sistemului în acest caz este:

$$X(t) = C_1 e^{\lambda t} v_1 + C_2 e^{\lambda t}(tv_1+v_2) = C_1 e^{4t}\begin{pmatrix}1\\1\end{pmatrix} + C_2 e^{4t}\begin{pmatrix}t+1\\t\end{pmatrix} = \begin{pmatrix}C_1 e^{4t} + C_2 e^{4t}(t+1)\\ C_1 e^{4t} + C_2 e^{4t}\cdot t\end{pmatrix}$$

Condiții inițiale: $\begin{cases} x(0)=0 \\ y(0)=1 \end{cases} \Rightarrow \begin{cases} C_1+C_2=0 \\ C_1=1 \end{cases} \Rightarrow \begin{cases} C_1=1 \\ C_2=-1 \end{cases}$

$\Rightarrow$ Soluția sistemului este:

$$\begin{cases} x(t) = e^{4t} - e^{4t}(t+1) = -te^{4t} \\ y(t) = e^{4t} - e^{4t}t = e^{4t}(1-t) \end{cases}$$

iar soluţia generală a sistemului $\mathbf{X}' = A\mathbf{X}$ va fi

$$\mathbf{X}(t) = P\begin{pmatrix} e^{\lambda t} & te^{\lambda t} \\ 0 & e^{\lambda t}\end{pmatrix}\begin{pmatrix}C_1\\C_2\end{pmatrix} = \underbrace{C_1 e^{\lambda t}\mathbf{v}_1}_{\mathbf{X}_1(t)} + \underbrace{C_2 e^{\lambda t}(t\mathbf{v}_1+\mathbf{v}_2)}_{\mathbf{X}_2(t)}$$

↳ caz omogen, A nediagonalizabilă

(c) $\begin{cases} x' = x-y \\ y' = -4x+y \end{cases}, x(0)=1, y(0)=-2$

$A=\begin{pmatrix}1 & -1\\ -4 & 1\end{pmatrix}$ $P_A(\lambda) = \begin{vmatrix} 1-\lambda & -1 \\ -4 & 1-\lambda\end{vmatrix} = (1-\lambda)^2 - 4 = \lambda^2 - 2\lambda + 1 - 4$

$= \lambda^2 - 2\lambda - 3 = (\lambda+1)(\lambda-3)$

Pentru $\lambda_1=-1 \Rightarrow \begin{cases} x-y=-x \\ -4x+y=-y \end{cases} \Leftrightarrow 2x=y \Rightarrow V_\lambda = Sp\{\begin{pmatrix}1\\2\end{pmatrix}\}$

$v_1=\begin{pmatrix}1\\2\end{pmatrix}$

Pentru $\lambda_2=3 \Rightarrow \begin{cases} x-y=3x \\ -4x+y=3y \end{cases} \Rightarrow y=-2x \Rightarrow V_\lambda = Sp\{\begin{pmatrix}1\\-2\end{pmatrix}\}$

$v_2=\begin{pmatrix}1\\-2\end{pmatrix}$

$\Rightarrow$ Soluția generală a sistemului dat este

$$X(t) = C_1 e^{\lambda_1 t} v_1 + C_2 e^{\lambda_2 t} v_2 = C_1 e^{-t}\cdot\begin{pmatrix}1\\2\end{pmatrix} + C_2 \cdot e^{3t}\cdot\begin{pmatrix}1\\-2\end{pmatrix} = \begin{pmatrix}C_1 e^{-t} + C_2 e^{3t} \\ 2C_1 e^{-t} - 2C_2 e^{3t}\end{pmatrix} \Rightarrow$$

$\begin{cases} x(0)=1 \\ y(0)=-2 \end{cases} \Rightarrow \begin{cases} c_1+c_2=1 \\ 2c_1-2c_2=-2 \end{cases} \Rightarrow \begin{cases} c_1+c_2=1 \\ c_1-c_2=-1 \end{cases} \Rightarrow 2c_1=0 \Rightarrow$

$\begin{cases} c_1=0 \\ c_2=1 \end{cases} \Rightarrow$ Soluția sistemului este $\begin{cases} x(t)=e^{3t} \\ y(t)=-2e^{3t} \end{cases}$

↳ cazul amuger. A diagonalizabilă

(d) $\begin{cases} x'=3x-y+z \\ y'=x+y+z \\ z'=4x-y+4z \end{cases}$ $x(0)=y(0)=z(0)=-2$

$A=\begin{pmatrix} 3 & -1 & 1 \\ 1 & 1 & 1 \\ 4 & -1 & 4 \end{pmatrix}$ $P_A(\lambda)=\begin{vmatrix} 3-\lambda & -1 & 1 \\ 1 & 1-\lambda & 1 \\ 4 & -1 & 4-\lambda \end{vmatrix}=$

$=-\lambda^3+8\lambda^2-17\lambda+10=$

$=-(\lambda-1)(\lambda-2)(\lambda-5)$

Pentru $\lambda_1=1 \Rightarrow \begin{cases} 3x-y+z=x \\ x+y+z=y \\ 4x-y+4z=z \end{cases} \Rightarrow \begin{cases} z=-x \\ y=2x+z=-2x+z=-x \end{cases} \Rightarrow$

$V_{\lambda_1}=Sp\{\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}\}$

Pentru $\lambda_2=2 \Rightarrow \begin{cases} 3x-y+z=2x \\ x+y+z=2y \\ 4x-y+4z=2z \end{cases} \Rightarrow \begin{cases} z=y-x \\ 4y-4x-y+4x=2z \end{cases} \Rightarrow$

$\begin{cases} x=y-z \\ 3y=2z \end{cases} \Rightarrow \begin{cases} x=y-\frac{3}{2}y=-1/2y \\ z=\frac{3}{2}y \end{cases} \Rightarrow V_{\lambda_2}=Sp\{\begin{pmatrix} -1/2 \\ 1 \\ 3/2 \end{pmatrix}\}=Sp\{\begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}\}$

Pentru $\lambda_3=5 \Rightarrow \begin{cases} 3x-y+z=5x \\ x+y+z=5y \\ 4x-y+4z=5z \end{cases} \Rightarrow V_{\lambda_3}=Sp\{\begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}\}$

$X'=PDP^{-1}X \Rightarrow P^{-1}X'=DP^{-1}X$

Notăm $P^{-1}X=\begin{pmatrix} u \\ v \\ w \end{pmatrix} \Rightarrow$ se obține sistemul de ecuații diferențiale decuplate.

$\begin{cases} u'=\lambda_1 u \\ v'=\lambda_2 v \\ w'=\lambda_3 w \end{cases} \Rightarrow \begin{pmatrix} u(t) \\ v(t) \\ w(t) \end{pmatrix}=\begin{pmatrix} c_1e^{\lambda_1 t} \\ c_2e^{\lambda_2 t} \\ c_3e^{\lambda_3 t} \end{pmatrix}=\begin{pmatrix} e^{\lambda_1 t} & 0 & 0 \\ 0 & e^{\lambda_2 t} & 0 \\ 0 & 0 & e^{\lambda_3 t} \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$

$$\Rightarrow X = P \begin{pmatrix} e^{\lambda_1 t} & 0 & 0 \\ 0 & e^{\lambda_2 t} & 0 \\ 0 & 0 & e^{\lambda_3 t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 2 & 1 \\ -1 & 3 & 3 \end{pmatrix} \begin{pmatrix} e^t & 0 & 0 \\ 0 & e^{2t} & 0 \\ 0 & 0 & e^{5t} \end{pmatrix}$$

$$\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} e^t & -e^{2t} & e^{5t} \\ e^t & 2e^{2t} & e^{5t} \\ -e^t & 3e^{2t} & 3e^{5t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} c_1 e^t - c_2 e^{2t} + c_3 e^{5t} \\ c_1 e^t - 2c_2 e^{2t} + c_3 e^{5t} \\ -c_1 e^5 + 3c_2 e^{2t} + 3c_3 e^{5t} \end{pmatrix}$$

$$\Rightarrow \quad x(0) = y(0) = z(0) = -2 \Rightarrow$$

$$\begin{cases} c_1 - c_2 + c_3 = -2 \\ c_1 - 2c_2 + c_3 = -2 \\ -c_1 + 3c_2 + 3c_3 = -2 \end{cases} \Rightarrow \begin{cases} c_1 + c_3 = -2 \\ c_2 = 0 \\ -c_1 + 3c_3 = -2 \end{cases} \Rightarrow \begin{cases} c_2 = 0 \\ c_3 = -1 \\ c_1 = -1 \end{cases} \Rightarrow$$

$$\begin{cases} x(t) = -e^t - e^{5t} \\ y(t) = -e^t - e^{5t} \\ z(t) = e^t - 3e^{5t} \end{cases}$$

Soluția sistemului

(e) $\begin{cases} x' = 3x - y - 3z \\ y' = -6x + 2y + 6z \\ z' = 6x - 2y - 6z \end{cases}, \quad x(0) = 1, \ y(0) = -1, \ z(0) = 0$

$$A = \begin{pmatrix} 3 & -1 & -3 \\ -6 & 2 & 6 \\ 6 & -2 & -6 \end{pmatrix} \qquad P_A(\lambda) = \begin{vmatrix} 3-\lambda & -1 & -3 \\ -6 & 2-\lambda & 6 \\ 6 & -2 & -6-\lambda \end{vmatrix} = -\lambda^3 - \lambda^2 = -\lambda^2(\lambda+1)$$

Pentru $\lambda_1 = 0 \Rightarrow \begin{cases} 3x - y - 3z = 0 \\ -6x + 2y + 6z = 0 \\ 6x - 2y - 6z = 0 \end{cases} \Leftrightarrow V_{\lambda_1} = Sp\{\underbrace{\begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}}_{v_1}, \underbrace{\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}}_{v_2}\}$

Pentru $\lambda_2 = -1 \Rightarrow \begin{cases} 3x - y - 3z = -x \\ -6x + 2y + 6z = -y \\ 6x - 2y - 6z = -z \end{cases} \Leftrightarrow V_{\lambda_2} = Sp\{\underbrace{\begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}}_{v_3}\}$

$$\Rightarrow X = (v_1 | v_2 | v_3) \begin{pmatrix} e^{0 \cdot t} & 0 & 0 \\ 0 & e^{0 \cdot t} & 0 \\ 0 & 0 & e^{-t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 3 & 0 & -2 \\ 0 & 1 & 2 \end{pmatrix}.$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{-t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & e^{-t} \\ 3 & 0 & -2e^{-t} \\ 0 & 1 & 2e^{-t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} c_1 + c_2 + e^{-t} c_3 \\ 3c_1 - 2c_3 e^{-t} \\ c_2 + c_3 \cdot 2e^{-t} \end{pmatrix}$$

$$\begin{cases} x(0)=1 \\ y(0)=-1 \\ z(0)=0 \end{cases} \Leftrightarrow \begin{cases} c_1+c_2+c_3=1 \\ 3c_1-2c_3=-1 \\ c_2+2c_3=0 \end{cases} \rightarrow \begin{cases} c_1=-3 \\ c_3=-4 \\ c_2=8 \end{cases} \Rightarrow$$

Soluția sistemului este $\begin{cases} x(t)=5-4e^{-t} \\ y(t)=-9+8e^{-t} \\ z(t)=8-8e^{-t} \end{cases}$

(f) $\begin{cases} x'=-4x+y \\ y'=-x-2y+e^{-3t} \end{cases}, x(0)=1, y(0)=1$

I Rezolvăm sistemul omogen asociat

$$\begin{cases} x'=-4x+y \\ y'=-x-2y \end{cases} \quad A=\begin{pmatrix} -4 & 1 \\ -1 & -2 \end{pmatrix} \quad P_A(\lambda)=\begin{vmatrix} -4-\lambda & 1 \\ -1 & -2-\lambda \end{vmatrix}=$$

$$=(4+\lambda)(2+\lambda)+1=8+4\lambda+2\lambda+\lambda^2+1=\lambda^2+6\lambda+9=$$

$$=(\lambda+3)^2$$

Pentru $\lambda=-3 \Rightarrow \begin{cases} -4x+y=-3x \\ -x-2y=-3y \end{cases} \Leftrightarrow V_\lambda=Sp\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\} \rightarrow$

A nu este diagonalizabilă

Fie $v_1=\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ și $v_2=\begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow$ soluția generală a sistemului este $X(t)=c_1e^{-3t}\begin{pmatrix} 1 \\ 1 \end{pmatrix}+c_2e^{-3t}\begin{pmatrix} t+1 \\ t \end{pmatrix}$

$$=\begin{pmatrix} c_1\cdot e^{-3t}+c_2e^{-3t}(t+1) \\ c_1e^{-3t}+c_2e^{-3t}\cdot t \end{pmatrix}$$

II Metoda variației constantelor

$$X(t)=\begin{pmatrix} e^{-3t} & e^{-3t}(t+1) \\ e^{-3t} & e^{-3t}t \end{pmatrix}\begin{pmatrix} c_1(t) \\ c_2(t) \end{pmatrix} \Rightarrow$$

$$\begin{pmatrix} e^{-3t} & e^{-3t}(t+1) \\ e^{-3t} & e^{-3t}\cdot t \end{pmatrix}\begin{pmatrix} c_1'(t) \\ c_2'(t) \end{pmatrix}=\begin{pmatrix} 0 \\ e^{-3t} \end{pmatrix} \Leftrightarrow$$

$$e^{-3t}\left(c_1'(t)+(t+1)\cdot c_2'(t)\right)=0$$

$$e^{-3t}\left(c_1'(t)+t\cdot c_2'(t)\right)=e^{-3t} \Leftrightarrow$$

$$\begin{cases} c_1'(t) + (t+1)\cdot c_2'(t) = 0 \\ c_1'(t) + t\cdot c_2'(t) = 1 \end{cases} \Rightarrow c_2'(t) = 1 \Rightarrow c_2(t) = t + k_2$$

$$c_1'(t) = -(t+1) \Rightarrow c_1(t) = -\frac{t^2}{2} - t + k_1 \Rightarrow$$

$$\begin{cases} x(t) = e^{-3t}\left(-\frac{t^2}{2} - t + k_1\right) + e^{-3t}(t+1)(t+k_2) \\ y(t) = e^{-3t}\left(-\frac{t^2}{2} - t + k_1\right) + e^{-3t}t(t+k_2) \end{cases}$$

$$\begin{cases} x(0) = 1 \\ y(0) = 1 \end{cases} \Rightarrow \begin{cases} k_1 + k_2 = 1 \\ k_1 = 1 \end{cases} \Leftrightarrow \begin{cases} k_1 = 1 \\ k_2 = 0 \end{cases} \Rightarrow$$

$$\begin{cases} x(t) = e^{-3t}\left(-\frac{t^2}{2} - t + 1\right) + e^{-3t}t(t+1) \\ y(t) = e^{-3t}\left(-\frac{t^2}{2} - t + 1\right) + e^{-3t}t^2 \end{cases} \Rightarrow$$

$$\begin{cases} x(t) = \frac{1}{2}e^{-3t}\left(-t^2 - 2t + 2 + 2t^2 + 2t\right) = \frac{1}{2}e^{-3t}(t^2+2) \\ y(t) = \frac{1}{2}e^{-3t}\left(-t^2 - 2t + 2 + 2t^2\right) = \frac{1}{2}e^{-3t}(t^2 - 2t + 2) \end{cases}$$

Metoda variaţiei constantelor

Propoziţie 2.16. Fie $\mathbf{X}_1, \ldots, \mathbf{X}_n : \mathbb{R} \to \mathbb{R}^n$ un sistem fundamental de soluţii pentru sistemul de ecuaţii diferenţiale liniare şi omogene $\mathbf{X}' = A\mathbf{X}$.
Atunci soluţia generală a sistemului neomogen $\mathbf{X}' = A\mathbf{X} + \mathbf{b}(t)$ este

$$\mathbf{X}(t) = C_1(t)\mathbf{X}_1(t) + \ldots + C_n(t)\mathbf{X}_n(t) = \mathbf{M}(t)\begin{pmatrix} C_1(t) \\ \vdots \\ C_n(t) \end{pmatrix}$$

unde $\mathbf{M}(t) = (\mathbf{X}_1(t) \mid \ldots \mid \mathbf{X}_n(t))$ este matricea fundamentală de soluţii, iar funcţiile $C_1, \ldots, C_n : I \to \mathbb{R}^n$ verifică relaţia

$$C_1'(t)\mathbf{X}_1(t) + \ldots + C_n'(t)\mathbf{X}_n(t) = \mathbf{M}(t)\begin{pmatrix} C_1'(t) \\ \vdots \\ C_n'(t) \end{pmatrix} = \mathbf{b}(t)$$

(g) $\begin{cases} x' = x + y - \cos t \\ y' = -2x - y + \sin t + \cos t \end{cases}$, $x(0) = 1$, $y(0) = -2$

I Rezolvăm sistemul omogen asociat

$$\begin{cases} x' = x + y \\ y' = -2x - y \end{cases} \quad A = \begin{pmatrix} 1 & 1 \\ -2 & -1 \end{pmatrix} \quad P_A(\lambda) = \begin{vmatrix} 1-\lambda & 1 \\ -2 & -1-\lambda \end{vmatrix} =$$

$$= -(1-\lambda)(1+\lambda) + 2 = \lambda^2 - 1 + 2 = \lambda^2 + 1$$

$\Rightarrow \lambda = \pm i$

Pentru $\lambda = i \Rightarrow \begin{cases} x + y = ix \\ -2x - y = iy \end{cases} \Rightarrow V_\lambda = Sp\{\overset{v}{\overbrace{\begin{pmatrix} -1-i \\ 2 \end{pmatrix}}}\}$

$\lambda = i$

$v = \underbrace{\begin{pmatrix} -1 \\ 2 \end{pmatrix}}_{v_1} + i\underbrace{\begin{pmatrix} -1 \\ 0 \end{pmatrix}}_{v_2} \Rightarrow$ Soluția generală a sistemului este

$$X(t) = (v_1 | v_2)\begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 2 & 0 \end{pmatrix}\begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$\begin{aligned} \mathbf{X}(t) &= C_1\mathbf{X}_1(t) + C_2\mathbf{X}_2(t) \\ &= C_1 e^{\alpha t}(\cos \beta t\, \mathbf{v}_1 - \sin \beta t\, \mathbf{v}_2) + C_2 e^{\alpha t}(\sin \beta t\, \mathbf{v}_1 + \cos \beta t\, \mathbf{v}_2) \\ &= e^{\alpha t}\,(\mathbf{v}_1 \mid \mathbf{v}_2)\begin{pmatrix} \cos \beta t & \sin \beta t \\ -\sin \beta t & \cos \beta t \end{pmatrix}\begin{pmatrix} C_1 \\ C_2 \end{pmatrix} \end{aligned} \tag{26}$$

↳ cazul omogen, A are valori proprii complexe

II Metoda variației constantelor

$$X(t) = \begin{pmatrix} -1 & -1 \\ 2 & 0 \end{pmatrix}\begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}\begin{pmatrix} c_1(t) \\ c_2(t) \end{pmatrix} \Rightarrow$$

$$\underbrace{\begin{pmatrix} -1 & -1 \\ 2 & 0 \end{pmatrix}\begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}}_{M(t)}\begin{pmatrix} c_1'(t) \\ c_2'(t) \end{pmatrix} = \begin{pmatrix} -\cos t \\ \sin t + \cos t \end{pmatrix} \Rightarrow$$

$$\begin{pmatrix} c_1'(t) \\ c_2'(t) \end{pmatrix} = (M(t))^{-1}\begin{pmatrix} -\cos t \\ \sin t + \cos t \end{pmatrix} \Rightarrow$$

$$\begin{pmatrix} c_1'(t) \\ c_2'(t) \end{pmatrix} = \begin{pmatrix} \sin t & \frac{\cos t + \sin t}{2} \\ -\cos t & \frac{-\cos t + \sin t}{2} \end{pmatrix}\begin{pmatrix} -\cos t \\ \sin t + \cos t \end{pmatrix} \Rightarrow$$

$$\begin{pmatrix} c_1'(t) \\ c_2'(t) \end{pmatrix} = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix} \Rightarrow \begin{cases} c_1(t) = \frac{t}{2} + k_1 \\ c_2(t) = \frac{t}{2} + k_2 \end{cases} \Rightarrow$$

$$X(t) = \begin{pmatrix} -1 & -1 \\ 2 & 0 \end{pmatrix}\begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}\begin{pmatrix} \frac{t}{2} + k_1 \\ \frac{t}{2} + k_2 \end{pmatrix}$$

$$\begin{cases} x(t) = (-\cos t + \sin t)\left(\frac{t}{2} + k_1\right) - (\cos t + \sin t)\left(\frac{t}{2} + k_2\right) \\ y(t) = 2\cos t\left(\frac{t}{2} + k_1\right) + 2\sin t\left(\frac{t}{2} + k_2\right) \end{cases}$$

$x(0) = 1,\ y(0) = -2 \Rightarrow \begin{cases} -k_1 - k_2 = 1 \\ 2k_1 = -2 \end{cases} \Rightarrow \begin{cases} k_1 = -1 \\ k_2 = 0 \end{cases}$

$\Rightarrow$ Soluția sistemului este:

$$\begin{cases} x(t) = \cos t\left(-\frac{t}{2}+1-\frac{t}{2}\right)+\sin t\left(\frac{t}{2}-\frac{t}{2}-1\right) = \cos t(-t+1)-\sin t \\ y(t) = \cos t\cdot t - 2\cos t + \sin t\cdot t \end{cases}$$

(h) $\begin{cases} x' = 2x+y+9t \\ y' = -x+4y \end{cases}$, $x(0)=0,\ y(0)=1$

I Rezolvăm sistemul omogen asociat

$$\begin{cases} x' = 2x+y \\ y' = -x+4y \end{cases} \quad A=\begin{pmatrix} 2 & 1 \\ -1 & 4 \end{pmatrix} \quad P_A(\lambda)=\begin{vmatrix} 2-\lambda & 1 \\ -1 & 4-\lambda \end{vmatrix} =$$

$$= (2-\lambda)(4-\lambda)+1 = \lambda^2-6\lambda+9 = (\lambda-3)^2$$

Pentru $\lambda=3 \Rightarrow \begin{cases} 2x+y=3x \\ -x+4y=3y \end{cases} \Leftrightarrow V_\lambda = Sp\{(1,1)\}$

Considerăm $v_1=\begin{pmatrix}1\\1\end{pmatrix}$, $v_2=\begin{pmatrix}0\\1\end{pmatrix}$ $\Rightarrow$ Soluția generală a sistemului este

$$X(t)=\begin{pmatrix}1&0\\1&1\end{pmatrix}\begin{pmatrix}e^{3t}&te^{3t}\\0&e^{3t}\end{pmatrix}\begin{pmatrix}c_1\\c_2\end{pmatrix}$$

II Metoda variației constantelor

$$\begin{pmatrix} e^{3t} & te^{3t} \\ e^{3t} & (t+1)e^{3t} \end{pmatrix}\begin{pmatrix} c_1'(t) \\ c_2'(t) \end{pmatrix} = \begin{pmatrix} 9t \\ 0 \end{pmatrix} \Rightarrow$$

$$\begin{pmatrix} c_1'(t) \\ c_2'(t) \end{pmatrix} = \begin{pmatrix} e^{3t} & te^{3t} \\ e^{3t} & (t+1)e^{3t} \end{pmatrix}^{-1}\begin{pmatrix} 9t \\ 0 \end{pmatrix} \Leftrightarrow \begin{pmatrix} c_1'(t) \\ c_2'(t) \end{pmatrix} =$$

$$\begin{pmatrix} (t+1)e^{-3t} & -te^{-3t} \\ -e^{-3t} & e^{-3t} \end{pmatrix}\begin{pmatrix} 9t \\ 0 \end{pmatrix} \Leftrightarrow \begin{pmatrix} c_1'(t) \\ c_2'(t) \end{pmatrix} = \begin{pmatrix} 9t(t+1)e^{-3t} \\ -9te^{-3t} \end{pmatrix}$$

$$\Leftrightarrow \begin{pmatrix} c_1(t) \\ c_2(t) \end{pmatrix} = \begin{pmatrix} \int 9t(t+1)e^{-3t}\,dt \\ \int -9te^{-3t}\,dt \end{pmatrix} \Leftrightarrow \begin{pmatrix} c_1(t) \\ c_2(t) \end{pmatrix} =$$

$$\begin{pmatrix} -\frac{(9t^2+15t+5)e^{-3t}}{3}+K_1 \\ (3t+1)e^{-3t}+K_2 \end{pmatrix}$$

$x(0)=0,\ y(0)=1$

$$\begin{cases} x(t) = e^{3t} c_1(t) + t \cdot e^{3t} c_2(t) \\ y(t) = e^{3t} c_1(t) + (t+1) e^{3t} c_2(t) \end{cases} \Rightarrow$$

$$\begin{cases} x(0) = c_1(0) \\ y(0) = c_1(0) + c_2(0) \end{cases} \Rightarrow \begin{cases} 0 = -\frac{5}{3} + K_1 \\ 1 = -\frac{5}{3} + K_1 + 1 + K_2 \end{cases}$$

$$\Rightarrow \begin{cases} K_1 = 5/3 \\ K_2 = 0 \end{cases} \Rightarrow$$

$$x(t) = -3t^2 - 5t - \frac{5}{3} + \frac{5}{3} \cdot e^{3t} + t(3t+1) = \frac{5}{3} e^{3t} - 4t - \frac{5}{3}$$

$$y(t) = \frac{5}{3} e^{-3t} - 4t - \frac{5}{3} + 3t + 1 = \frac{5}{3} e^{3t} - t - \frac{2}{3}$$